

Hrigved Suryawanshi

✉ suryawanshi.h@northeastern.edu

☎ +1 (857) 675-0700

🌐 Portfolio

🌐 LinkedIn

EDUCATION

Northeastern University, Boston, MA

Sep 2023 – May 2025

Master of Science in Robotics (with [THESIS](#))

Relevant Coursework: Pattern Recognition, Computer Vision, Mobile Robotics, Sensing & Navigation, Robot Mechanics

University of Mumbai, Mumbai, India

Aug 2019 - May 2023

Bachelor of Engineering in Electronics and Telecommunications

Honors: Best Innovator Award for Sustainable [Electric Motorcycle](#) Project

PROFESSIONAL EXPERIENCE

[Agrobotics](#) | Robotics Autonomy and Software Engineer | Salinas, CA

Jan 2026 - Present

- Developing AR-based and Aloha style **teleoperation pipeline** using Meta Quest, Vision Pro and custom exoskeleton on xArm/OpenArms manipulators, enabling low-latency manipulation tasks handling gravity compensation, velocity smoothing and PID control.
- Architecting **LiDAR-stereo fusion** SLAM and NAV2 navigation pipeline for autonomous field robots, integrating point cloud and visual odometry for robust localization in GPS-degraded outdoor environments
- Designing custom **behavior tree** framework for mission planning and task sequencing, enabling modular state transitions between autonomous navigation, manipulation, and teleoperation modes
- Fine-tuning **VLA models** (OpenVLA, π_0) for vision-language-action grasping, bridging perception and manipulation for autonomous harvesting workflows
- Configuring Motorcortex and Intel NUC with embedded Linux, integrating sensors and actuators over CAN, SPI, I2C, and UART; set up network infrastructure including Ethernet switches and AP communication for real-time robot telemetry and control.

[Silicon Synapse Lab](#) | Research Assistant | Boston, MA

Jun 2024 - Dec 2025

M4 (Multimodal Morphing Robot):

- Developed **ROS2** decision node leveraging CNN traversability estimator for optimal ground/aerial mode selection, integrating terrain roughness, slope, and battery efficiency into custom **NAV2** costmaps (89% accuracy); engineered elevation mapping fusing Livox LiDAR with RealSense D455 (0.05m resolution, 100Hz)
- Designed NAV2/MAVROS integration with DWB local planner, enabling 6-second morphing transitions based on traversability scores.

Cobra (Snake Robot for Martian Terrain):

- Benchmarked **SLAM** algorithms (LIO-SAM vs RTAB-Map), achieving 2cm localization accuracy validated against OptiTrack; reduced mapping error by 25% through sensor calibration
- Performed point cloud segmentation using **PCL** for terrain analysis and CloudCompare for temporal change detection, processing 100k+ points/frame to identify navigable surfaces with 94% accuracy

FYNDME | Deep Learning Engineer Intern | Boston, MA

Jan 2025 - May 2025

- Improved face recognition accuracy from 60% to 85% by implementing dual-layer anti-spoofing with **InsightFace** and DeepPixBIS, reducing false acceptance rate to 2% for secure authentication
- Achieved 92% mAP@0.5 for real-time SKU detection using **YOLOv8** across 500+ product categories, decreasing product search time for delivery drivers navigating unfamiliar retail environments

VHIRON | Embedded Systems Engineer | Mumbai, India

Oct 2022 - Mar 2023

- Developed 3D photogrammetry scanner achieving 0.1mm resolution using Raspberry Pi and motorized turntable, processing 1M+ point clouds through cloud pipeline with 50% time reduction
- Constructed 18-inch CNC laser cutter with 0.05mm precision using GRBL firmware and ran G-code optimization for rapid prototyping applications

TECHNICAL SKILLS

Programming: Python, C/C++, Bash, MATLAB, Linux

Robotics: ROS/ROS2, Gazebo, MoveIt, NAV2, MAVROS, PCL, SLAM (ORB-SLAM3, RTAB-Map, LIO-SAM, FAST-LIO)

ML/CV: OpenVLA, OpenPI, YOLO, MobileNet, InsightFace, DeepPixBIS, CLIP,

Hardware: Jetson, Pixhawk, LiDAR, RealSense, RPi, Arduino, xArm, Openarms

Tools: Docker, Git, MeshLab, CloudCompare, Solidworks, MATLAB Simulink, Colmap, PyBullet, MuJoCo

PROJECTS

Autonomous AI Drone Platform

May 2024 - Present

- Architected fully autonomous UAV with **Pixhawk PX4** and Jetson Orin, implementing Extended Kalman Filter for 9-axis IMU.
- Integrated Orbbec Gemini 335L depth camera for obstacle avoidance (2m clearance, 100ms reaction) and custom PID-based human tracking maintaining 3m following distance with 0.5m position accuracy
- Developed ArUco-based precision landing achieving 5cm accuracy and deployed **RTAB-Map** SLAM for real-time 3D mapping

PUBLICATIONS

- **Simple User-Friendly MIDI Controller using Arduino**
- **ECG-based Arrhythmia Detection using CNN**